Assignment – 1.20

Firewall Basics Theory

Q - 1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

- **Block Spotify (Outbound Rule):**

1. Press Win + R, type wf.msc, and press Enter.
2. Click **Outbound Rules → New Rule**.
3. Select **Program** and click Next.
4. Browse and select Spotify.exe.
5. Select **Block the connection**.
6. Select all profiles: Domain, Private, and Public.
7. Name the rule **Block Spotify** and click Finish.

Q - 2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

- **Allow Google Chrome on Port 443 (Inbound Rule):**

1. Open wf.msc.
2. Click **Inbound Rules → New Rule → Custom**.
3. Select the Chrome program path.
4. Select **TCP** and enter local port 443.
5. Select **Allow the connection**.
6. Select the required profiles.
7. Name the rule **Allow Chrome Port 443** and click Finish.

Q - 3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

- **Block a Specific IP Address:**

  Example IP: 123.45.67.89

  1. Open wf.msc.
  2. Select **Inbound Rules → New Rule → Custom**.
  3. Select **All programs** and protocol **Any**.
  4. Open the **Scope** tab.
  5. Under **Remote IP address**, select **These IP addresses → Add**.
  6. Enter 123.45.67.89 and click OK.
  7. Select **Block the connection**.
  8. Select all required profiles.
  9. Name the rule **Block IP Address** and click Finish.

Q - 4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

➢ Hardware Firewall vs. Software Firewall

Hardware Firewall	Software Firewall
1. A physical device.	1. A program on a computer.
2. Protects many devices on a network.	2. Mainly protects one computer.
3. Used in offices and companies.	3. Used on home and office computers.
Use case: Protecting an office network.	**Use case:** Blocking unwanted apps on a laptop.